

SECURING STATE REVENUE THROUGH BLOCKCHAIN: THE MEDIATING ROLE OF PERCEIVED TRUST IN TELECOMMUNICATIONS IT GOVERNANCE

Abstract

Deploying blockchain in Indonesia's telecommunications fine administration promises enhanced transparency but faces hurdles from institutional complexity and trust deficits. Addressing the gap in IT governance literature, this study investigates public managers' behavioral intention to use (BIU) blockchain, testing the Trust-Guided Technology Assimilation (TGTA) mechanism where Perceived Trust (PT) mediates Technology Acceptance Model (TAM) constructs. Using a quantitative approach, data from 146 managers at Radio Monitoring Office Regional Units were analyzed via PLS-SEM. The model explains 54.2% of the variance in BIU. While both Perceived Usefulness (PU) and Perceived Ease of Use (PEU) directly drive BIU, PU exerts the strongest effect on PT ($\beta = 0.548$), demonstrating that functional benefits primarily engineer institutional trust. Crucially, PT partially mediates the PU/PEU-to-BIU relationships, validating the TGTA mechanism. The findings establish that blockchain adoption in high-risk governance is a psychological reconstruction rather than a linear technical upgrade, providing strategic insights for IT governance and financial management in developing economies.

Keywords: Blockchain adoption; Perceived trust; IT Governance; Technology Acceptance Model; Behavioral intention

INTRODUCTION

The scholarly inquiry into the implementation of blockchain technology (BCT) has exerted a profound influence across diverse domains, notably the public service sector, owing to its capacity to transform operational methodologies, enhance transparency, and mitigate fraudulent activities (Norbu et al., 2024). Digital governance requires frameworks that facilitate accountability, delineate roles, and define decision-making authority. In this regard, BCT presents a viable solution through its decentralized, immutable ledger system, which facilitates transparent and traceable transactions (Shaheen et al., 2025). Emerging technologies, particularly distributed ledger technologies (DLT), significantly aid in fraud detection and auditability within financial and public ecosystems (Colline, 2025). However, while the adoption of BCT in the private sector (e.g., e-commerce, supply chain) has been widely explored, its implementation in the public sector remains complex, particularly concerning user adoption by civil servants who must transition from manual, control-based workflows to decentralized systems.

The global enthusiasm for BCT governance contrasts sharply with the hesitant adoption trajectories observed in developing economies, particularly within the ASEAN region. Countries like Indonesia face a unique confluence of challenges, including fragmented digital infrastructure, acute digital skills gaps, and institutional resistance to change (Taherdoost, 2022). Previous studies have delineated a notable neglect regarding the utilization of DLT in developing countries, revealing a substantial research deficit (Ullah et al., 2022). Although blockchain has attracted significant academic attention, there is a glaring empirical gap regarding its adoption in developing countries, particularly within the specific context of high-risk public services.

Indonesia, as a rapidly digitizing economy where non-tax state revenue (PNBP) relies heavily on compliance sectors, provides a critical empirical context. A key area requiring modernization is the management of administrative fine sanctions within the telecommunications sector. This sector contributes significantly to national revenue but remains prone to procedural human errors and fraud loopholes due to incomplete digital automation. The current mechanism for managing administrative fines—involving multi-layered supervision and sensitive data exchanges—suffers from organizational silos and operational opacity. Adopting BCT in this domain directly addresses the

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need for secure, traceable transactions, thereby enhancing accountability and public confidence (Ahmad & Shah, 2022).

However, deploying BCT in highly regulated public entities, such as the Radio Monitoring Office Regional Unit, is inherently complex. It occurs within an environment of Institutional Complexity—characterized by rigid bureaucratic structures, hierarchical compliance requirements, and deeply entrenched cultural resistance among civil servants. In such high-risk settings, technology adoption is not merely a technical upgrade but a reconstruction of institutional confidence (figure 1.).



Figure. 1 Research gap

This reality underscores the limitations of applying generic technology adoption frameworks, which often fail to account for the unique pressure points inherent to the bureaucratic structures of developing countries. Three critical research gaps emerge. First, the decontextualization of trust,

where traditional **Technology Acceptance Model (TAM)** neglects **Perceived Trust (PT)** PT as a prerequisite in high-risk sectors (Belanche et al., 2012). Second, institutional complexity, wherein standard theories overlook bureaucratic pressures (Ullah et al., 2022). Third, an empirical deficiency in governance, as literature predominantly focuses on commercial supply chains, leaving a glaring gap in DLT adoption among public service personnel (Bano & Siddiqui, 2025; Taherdoost, 2022).

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To address these gaps, this study proposes a hybrid model integrating the core constructs of the Technology Acceptance Model (TAM)— **Perceived Usefulness (PU)** and **Perceived Ease of Use (PEU)** PU and PEU—with the Theory of Planned Behavior (TPB), enriched by Perceived Trust (PT) as a mediating variable. While classical TAM posits that utility and ease of use directly determine behavioral intent, we argue that in the context of public fine sanction enforcement, these functional perceptions must first be internalized as trust in the system's reliability. Trust acts as a psychological bridge, mitigating the perceived risks associated with transitioning from traditional to decentralized systems.

Therefore, this study is guided by the following Research Questions (RQ):

RQ1: What are the direct effects of Perceived Usefulness (PU) and Perceived Ease of Use (PEU) on the Behavioral Intention to Use (BIU) blockchain technology for sanctions management?

RQ2: Does Perceived Trust (PT) mediate the relationship between TAM constructs (PU, PEU) and BIU in the context of high-risk public sector governance?

Accordingly, the Research Objectives (RO) are:

RO1: To evaluate the direct effects of PU and PEU on BIU blockchain for sanctions management among civil servants.

RO2: To assess the mediating role of PT in the relationship between TAM constructs (PU, PEU) and BIU, thereby testing the Trust-Guided Technology Assimilation (TGTA) mechanism.

By answering these questions, this study provides pivotal theoretical and practical contributions. Theoretically, it advances TAM-TPB literature by empirically validating Perceived Trust as a cognitive filter (mediator) in a high-risk bureaucratic setting, moving beyond the traditional linear adoption model. Practically, as Indonesia accelerates its digital transformation, the findings will provide a strategic roadmap for the Ministry of Communication and Digital (Komdigi) in formulating implementation policies that prioritize not only system utility and convenience but, most importantly, the cultivation of user trust to ensure successful blockchain integration.

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https://www.researchgate.net/publication/359945936_The_Intention_to_Use_Blockchain_in_Indonesia_Using_Extended_Approach_Technology_Acceptance_Model_TAM

https://www.researchgate.net/publication/389251381_Legal_Regulations_on_The_Use_of_Blockchain_Technology_in_Indonesia

https://www.researchgate.net/publication/377195001_Implementation_of_Blockchain_Technology_in_Improving_Transparency_of_Public_Services_A_Case_Study_on_Government_Service_Delivery_in_Indonesia

You can find more here

<https://share.google/aimode/VbDenIJVSFwjFFGT>

2) similar hypothesis studies testing acceptance (including those in 1) and maybe comparing results with the factor of developing /developed countries.

LITERATURE REVIEW AND HYPOTHESIS DEVELOPMENT

This is more of defining terminology and hypothesis development; there is no literature survey here. Better call it something like “Hypothesis development & Performance Metrics used” or just “Hypothesis development” and move the 1st two lines defining each term to the introduction

Technology Acceptance Model (TAM)

The TAM is widely recognized as the predominant theoretical framework for elucidating user acceptance of novel technologies. TAM accentuates two fundamental constructs: PU and PEU. PU denotes the degree to which a technology is perceived to enhance job performance, while PEU represents the extent to which the technology is regarded as user-friendly and requires minimal cognitive effort to operate. In the context of this study, PU encompasses the degree to which civil servants perceive blockchain as a means to enhance efficiency, transparency, and accountability in the management of telecommunications fines, as well as to reduce the likelihood of fraudulent activities. PEU encompasses how straightforward respondents find the operation of the blockchain system, including considerations of required training, user interface design, and integration with pre-existing bureaucratic workflows. While TAM is robust in explaining utilitarian and ergonomic drivers of adoption, its traditional form overlooks the psychological and institutional barriers inherent in high-risk public sector environments.

Theory of Planned Behavior (TPB) and Behavioral Intention

To address the limitations of TAM in explaining volitional behavior in complex organizational settings, this study integrates the (Abubker et al., 2021). TPB posits that BIU is the most accurate predictor of actual behavior, formed by attitudes, subjective norms, and perceived behavioral control. In the context of adopting blockchain for fine sanctions management, BIU reflects the degree of intent or willingness among civil servants to regularly utilize the new decentralized system. While TAM explains the technological perception, TPB provides the framework for the behavioral intention, acknowledging that adoption in government institutions is subject to organizational pressures and individual agency.

Perceived Trust (PT) as a Cognitive Filter

Although TAM and TPB provide foundational explanatory power, their traditional formulations neglect a critical prerequisite for technology adoption in the public sector: Trust. Perceived Trust (PT) encompasses the conviction that a technology is dependable, secure, and aligned with the interests of its users (Rahardja et al., 2023). In the domain of public fine sanction enforcement—where data immutability and auditability are paramount—civil servants must trust that the system will not erroneously record data or expose sensitive information. Drawing upon Social Belief Theory and the Uncertainty Reduction Model, we argue that PT acts as a "cognitive filter" or a mechanism for technological legitimization. It transforms the uncertainty of a novel disruptive technology into institutional belief, building a psychological bridge between the individual user and the system. Consequently, integrating PT as a mediating variable bridges the gap between functional perceptions (PU/PEU) and behavioral outcomes (BIU).

Blockchain Technology Adoption and Hypothesis Development

The integration of blockchain technology into public governance frameworks is instrumental in combating corruption and enhancing auditability (Ayebofo et al., 2025). However, for civil servants at the Radio Monitoring Office Regional Unit, transitioning from traditional, manual oversight to a trust-minimized decentralized ledger represents a significant psychological shift. Figure 2 illustrates the conceptual model and the hypothesized relationships among TAM, TPB, and Perceived Trust within the context of the Radio Monitoring Office Regional Unit.

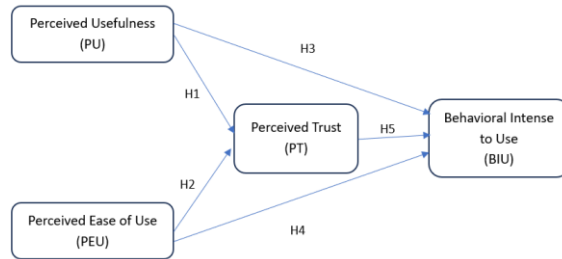


Figure. 2 Conceptual model

Based on the theoretical synthesis of TAM, TPB, and Trust literature, we propose the following hypotheses:

H1: Perceived Usefulness (PU) has a significant positive effect on Perceived Trust (PT)

Trust in innovative technologies within bureaucratic settings is predominantly driven by the manifestation of their functional capabilities. According to Attribution Theory and the Cognitive Evaluation Model, when users perceive that a technology provides tangible benefits—such as absolute transparency, decentralized data security, and reduced bureaucracy—they attribute credibility to that system. In the context of high-risk fine management, if civil servants believe blockchain improves their performance (PU), they are more likely to trust its reliability (PT), creating a virtue circle between utility and trust.

H2: Perceived Ease of Use (PEU) has a significant positive effect on Perceived Trust (PT)

When a blockchain system is perceived as PEU, respondents are inclined to regard it as more reliable because achieving satisfactory outcomes requires minimal effort (Tian, 2024). A complex, cumbersome system breeds skepticism regarding its dependability. Therefore, the easier the blockchain system is to operate, the higher the civil servant's confidence in its technical reliability.

H3: Perceived Usefulness (PU) exerts a significant positive influence on Behavioral Intention to Use (BIU)

Within the paradigm of fine-sanctions management, if respondents believe that blockchain technology will markedly enhance the accuracy, efficiency, and transparency of the sanctions enforcement mechanism, their intention to use it (BIU) will consequently increase. Driven by Expectation-Value Theory, individuals evaluate technology based on its potential benefits; thus, higher perceived utility translates to stronger adoption intention.

H4: Perceived Ease of Use (PEU) exerts a significant positive influence on Behavioral Intention to Use (BIU)

PEU is articulated as the extent to which individuals perceive the utilization of a novel system as devoid of effort. Although the impact of PEU is frequently overshadowed by PU, in bureaucratic settings characterized by institutional complexity, a user-friendly system is imperative to mitigate resistance to change.

H5: Perceived Trust (PT) exerts a significant positive influence on Behavioral Intention to Use (BIU)

Research has established that incorporating Trust into technology adoption models is essential to mitigate uncertainties in digital contexts (Belanche et al., 2012). Trust acts as a catalyst; without the belief that the system is secure and dependable, civil servants will hesitate to adopt it, regardless of its utility.

H6: Perceived Trust (PT) mediates the effect of Perceived Usefulness (PU) on Behavioral Intention to Use (BIU)

Building on the TGTA mechanism, functional benefits (PU) must be internalized as psychological assurance. The perceived advantages of blockchain (e.g., data traceability) must first be translated

into a sense of confidence (PT) in the system's security before civil servants are willing to commit to using it (BIU). PT functions as a vital conduit through which utility drives adoption.

H7: Perceived Trust (PT) mediates the effect of Perceived Ease of Use (PEU) on Behavioral Intention to Use (BIU)

H6 & H7 are not shown in the diagram; regarding their meaning and the lines below, you may call them H5a & H5b or H5pu & H5peu or add them to the figure in place of H5.

The perception that a system is easy to use reduces the fear of operational failure, thereby fostering trust. This trust, in turn, drives the behavioral intention to adopt. This mediation pathway underscores that convenience must be internalized as system reliability to effect behavioral change in risk-averse public sector environments.

METHODS

Research Design and Context

This study employs a quantitative-explanatory approach to test the causal relationships within the proposed hybrid TAM-TPB-Trust model. The research context is the management of administrative fine sanctions in Indonesia's telecommunications sector, a high-stakes governance domain requiring immutable audit trails and multi-stakeholder accountability. Data collection targeted civil servants at the Radio Monitoring Office Regional Units across 38 provinces in Indonesia, managed by the Ministry of Communication and Digital (Komdigi).

Population and Sampling

Information on the response rate and demographic characteristics of Civil Servants at the Monitoring Center of the Ministry of Communication and Digital of the Republic of Indonesia, spread across all regional offices throughout Indonesia, consisting of 38 provinces, as depicted in Figure 3. Respondents included Heads of Office, Work Team Leaders at the managerial level, Persons in Charge (PICs) of telecommunications fines, field implementers, and investigators. The population for this research encompasses employees from governmental agencies and users of the radio frequency administrative sanction payment system who are directly engaged in this process.



Figure. 3 Respondents demographics

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The target population comprises civil servants who are directly involved in the administration of telecommunications fine sanctions (e.g., Heads of Office, IT/system managers, field inspectors, and investigators). Due to the specific criteria required for the study, a purposive sampling technique was employed. The inclusion criteria were: (1) respondents must be users or administrators of the current

administrative sanction payment system, and (2) respondents must possess a fundamental understanding of digital technology concepts.

From a total population of 160 eligible civil servants, 146 valid responses were collected, yielding a response rate of 91.25%. The sample size of 146 exceeds the minimum threshold of 100–200 participants recommended for **Partial Least Square Structural Equation Modeling (PLS-SEM)** with complex models involving mediating variables (Hair et al., 2019). Furthermore, it satisfies the more rigorous "10-times rule" (requiring a minimum of 50 samples for 5 structural paths) and GPower requirements for a medium effect size at a 95% confidence level. Stratification was ensured to capture representation across managerial, technical, and operational hierarchies. **Define GPower abbreviation also and add a ref. for it.**

Data Collection, Measures and Instrument Development

Data were collected via an online structured questionnaire hosted on Google Forms between July and November 2025.

The instrument was developed by adapting validated scales from established technology adoption literature and contextualizing them for the Indonesian telecommunications 2s domain. All items were measured using a 5-point Likert scale, ranging from 1 (Strongly Disagree) to 5 (Strongly Agree). To ensure linguistic equivalence, the items underwent a rigorous back-translation process from English to Bahasa Indonesia. A pilot test was conducted on 30 civil servants (excluded from the main sample) to confirm semantic clarity and face validity.

The operationalization of the constructs—aligned with the Research Objectives (RO1 and RO2)—is detailed in Table 1.

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- 2-The population of workers the sample was taken from, does it reflect the same ratios in the sample especially regarding the education level?

Table 1. Construct operationalization and measurement items

Construct	Definition	Indicators	Source Adapted
Perceived Usefulness (PU)	The degree to which a civil servant believes blockchain enhances the efficiency and transparency of sanctions administration.	PU1: The use of Blockchain Technology can increase the transparency of the management of fine sanctions.	(Ayebofo et al., 2025) (Ullah et al., 2021) (Taherdoost, 2022) (Sasitharan et al., 2023)
		PU2: The use of Blockchain Technology can increase the traceability of data recording of fines.	
		PU3: The use of Blockchain Technology will save me time and effort in the tasks of managing fine sanctions.	
		PU4: The use of Blockchain Technology will improve the quality of the decisions I make regarding the sanction of fines.	
		PU5: I expect the features of Blockchain Technology to be easy to understand.	
Perceived Ease of Use (PEU)	The extent to which a civil servant perceives blockchain	PEU1: I feel that Blockchain Technology is easy to learn.	(Ayebofo et al., 2025) (Ullah et al., 2021)
		PEU2: I feel that I will be able to easily become proficient in using a fine sanctions	

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Construct	Definition	Indicators	Source Adapted
Perceived Trust (PT)	as easy to learn and operate.	management system based on Blockchain Technology. PEU3: I think Blockchain Technology will be easier to use than the current fine sanctions management process. PEU4: I am excited about the idea of adopting Blockchain Technology in the management of fine sanctions. PT1: I have confidence in the reliability of Blockchain Technology solutions for fine sanctions management.	(Taherdoost, 2022) (Sasitharan et al., 2023)
	The institutional and system-based confidence that blockchain will not be misused for surveillance or punitive retroactivity.	PT2: I believe that Blockchain Technology can effectively maintain the security of fine-sanctioned data even without continuous surveillance. PT3: I believe that the data stored in Blockchain Technology will not be easily manipulated. PT4: I have enough knowledge and skills to use Blockchain Technology in the management of fine sanctions.	(Toader et al., 2024) (Norbu et al., 2024)
	Behavioral Intention Use (BIU) The likelihood that a civil servant will adopt blockchain for sanctions tasks.	BIU1: I intend to use Blockchain Technology if it is available for the management of fine sanctions.	(Ullah et al., 2022)
		BIU2: I will voluntarily utilize Blockchain Technology for fine sanctions management tasks. BIU3: I would recommend the use of Blockchain Technology for the management of fine sanctions to my colleagues.	(Ullah et al., 2021)

Note: Items were adapted to fit the specific context of blockchain adoption in public sector sanctions enforcement.

Data Analysis Technique

The stages of analysis include the following: Outer Model, Convergent Validity, Discriminant Validity, Construct Reliability, Inner Model, Hypothesis Testing, Path Coefficients, and Significance (Figure.4). Hypothesis testing uses the following stages: Bootstrapping, t-statistic, p-value, and 95% Confidence Interval. The data analysis was conducted using PLS-SEM in SmartPLS 3.0.

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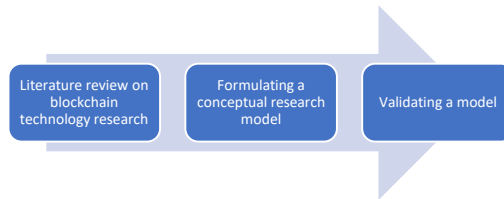


Figure. 4 Research steps

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PLS-SEM was chosen over CB-SEM for three methodologically justified reasons: (1) the primary objective of the study is prediction and theory development (explaining variance in BIU) rather than confirmatory theory testing; (2) the model incorporates complex mediating pathways which PLS-SEM handles robustly; and (3) the sample size (N=146) aligns optimally with PLS-SEM requirements for predictive models (Ramalho et al., 2019; Ullah et al., 2021).

The data analysis followed a rigorous two-step approach designed to directly answer the proposed Research Questions:

Step 1: Measurement Model Assessment (Outer Model)

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This step ensures the reliability and validity of the constructs before assessing structural relationships.

- Convergent Validity: Evaluated through outer loadings (threshold > 0.708) and Average Variance Extracted (AVE > 0.50).
- Discriminant Validity: Assessed using the HTMT ratio (threshold < 0.90), which is the modern standard for PLS-SEM, supplemented by the Fornell-Larcker criterion. This ensures that PT is empirically distinct from PU and PEU—a strict prerequisite for testing mediation (RO2).
- Internal Consistency Reliability: Measured via Composite Reliability (CR > 0.70) and Cronbach's Alpha (> 0.70).
- Common Method Bias (CMB): Addressed procedurally through respondent anonymity and counterbalanced question order, and statistically via Harman's single-factor test (first factor variance must be < 50%).

Step 2: Structural Model Assessment (Inner Model)

Once the measurement model is validated, the structural model is assessed to test the hypotheses (H1–H7) and answer the RQs.

- Collinearity: Assessed using Variance Inflation Factor (VIF), where values < 5 indicate no critical collinearity issues.
- Direct Effects (Answering RQ1 / Testing H1–H5): Path coefficients (β) are estimated using the ordinary least squares algorithm. Significance is assessed via bootstrapping with 5,000 subsamples to generate t-statistics and p-values (two-tailed, $\alpha = 0.05$). This directly identifies the main drivers of PT and BIU.
- Effect Sizes (f^2): Calculated to assess practical significance (0.02 = small, 0.15 = medium, 0.35 = large).
- Mediation Analysis (Answering RQ2 / Testing H6–H7): Specific indirect effects are calculated via bootstrapping. If the indirect path (e.g., PU → PT → BIU) is significant, and the direct path remains significant, partial mediation is established. This explicitly answers how PT functions as a cognitive filter.

- Predictive Power and Relevance: The coefficient of determination (R^2) explains the variance in endogenous variables (PT and BIU). Predictive relevance (Q^2) is calculated using blindfolding ($Q^2 > 0$ indicates predictive relevance), ensuring the model's utility for policy formulation.
- Model Fit: Evaluated using the Standardized Root Mean Square Residual (SRMR < 0.08) and Normed Fit Index (NFI).

RESULTS

Respondent Profile

A total of 146 valid responses were collected from civil servants at the Radio Monitoring Office Regional Units across Indonesia. The demographic profile indicates a highly experienced and well-educated workforce, which is appropriate for assessing complex technological adoption (table 2). The majority of respondents hold a Master's degree (42.47%) or Bachelor's degree (39.04%). Furthermore, 78.08% of the respondents possess more than 10 years of work experience, and 52.05% serve in technical functional roles, with 15.75% acting as law officers (PPNS). This demographic composition ensures that the respondents possess the requisite procedural knowledge and operational understanding of the current fine sanctions management system to evaluate the proposed blockchain technology reliably.

Table 2. Respondent profile

Respondent characteristic	Response variants	Percentage
Education level	High school/vocational	2.75
	Diploma 3	15.75
	Bachelor	39.04
	Magister	42.47
Work Experience	1 – 5 years	6.16
	6 – 10 years	15.75
	> 10 years	78.08
The Role of Positions	Regional head of office	10.27
	Team leader	21.92
	Technical functional	52.05
	PPNS	15.75
Frequency of interaction with the telecommunications sector with the administrative sanctions system	Several times a week	5.04
	Several times a month	27.34
	Several time a year	51.08
	Very rare	16.55

Measurement Model Assessment (Outer Model)

Prior to evaluating the structural relationships, the measurement model was assessed for reliability and validity to ensure the robustness of the constructs (figure 5).

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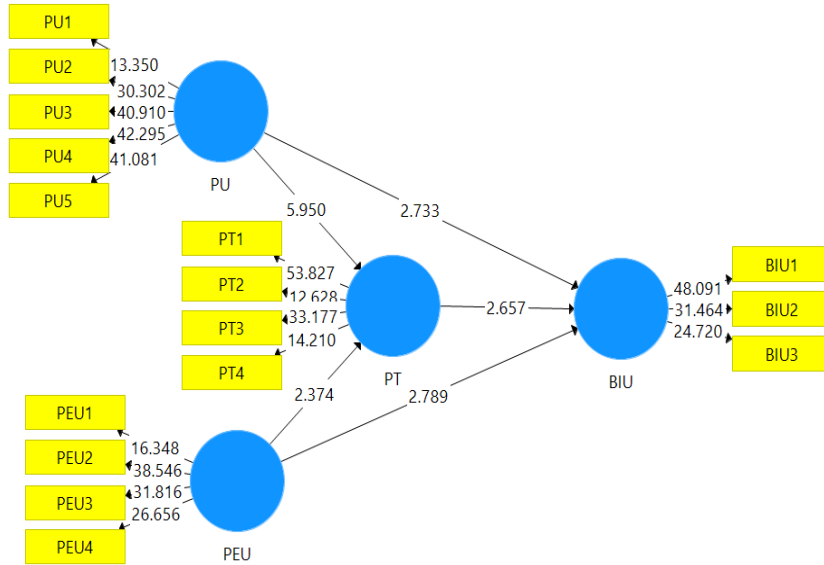


Figure. 5 Smart PLS Result

As presented in Table 3, all outer loadings exceeded the 0.708 threshold, indicating strong individual indicator reliability. Internal consistency was confirmed, with Cronbach's Alpha values ranging from 0.864 to 0.916 and Composite Reliability (CR) values ranging from 0.907 to 0.937, all surpassing the 0.70 criterion. Convergent validity was established, as the AVE for all constructs (PU, PEU, PT, BIU) exceeded the 0.50 threshold, confirming that each construct explains more than half of the variance of its indicators.

Table 3. Outer model

Construct	Outer Loadings (Min.)	Cronbach's Alpha	Composite Reliability (CR)	Average Variance Extracted (AVE)	Criterion
BIU	0.862	0.872	0.921	0.796	Pass (CR>0.7, AVE>0.5)
PEU	0.813	0.880	0.918	0.736	Pass (CR>0.7, AVE>0.5)
PT	0.769	0.864	0.907	0.710	Pass (CR>0.7, AVE>0.5)
PU	0.792	0.916	0.937	0.749	Pass (CR>0.7, AVE>0.5)

Discriminant validity was assessed using the HTMT (define abbreviation) ratio. As shown in Table 4, all HTMT values are below the conservative threshold of 0.90, supplemented by the Fornell-Larcker criterion (Table 5) where the square root of AVE for each construct exceeds its highest correlation with any other construct. This empirically confirms that PT is distinct from PU and PEU—a crucial prerequisite for testing the mediation model (RO2).

Table 4. HTMT check

BIU	FEU	PT	PU
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BIU			
PEU	0.703		
PT	0.733	0.678	
PU	0.734	0.721	0.785

Table 5. Fornell-Larcker criterion

	BIU	PEU	PT	PU
BIU	0.892			
PEU	0.623	0.858		
PT	0.653	0.602	0.843	
PU	0.662	0.657	0.707	0.865

Structural Model Assessment (Inner Model) and Hypothesis Testing

After establishing the measurement model's validity, the structural model was evaluated to test the proposed hypotheses (H1–H7) and address the Research Questions. Collinearity issues were ruled out as all VIF values were well below the threshold of 5. The structural model demonstrates a good fit, with a Standardized Root Mean Square Residual (SRMR) of 0.071 (< 0.08) and a Normed Fit Index (NFI) of 0.811.

Direct Effects (Addressing RQ1 / RO1)

RQ1 asks about the direct effects of PU and PEU on BIU, as well as their impact on PT. The bootstrapping results (5,000 subsamples) for the direct pathways are summarized in Table 6.

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Table 6. The hypothesis test (direct path)

Path Hypothesis	Line Coefficient (β)	T-Statistic	P-Value	Results	Effect Size (f^2)
H1: PU $\rightarrow$ PT	0.548	5.950	0.000	Supported	0.365 (Strong)
H2: PEU $\rightarrow$ PT	0.242	2.374	0.018	Supported	0.075 (Medium)
H3: PU $\rightarrow$ BIU	0.281	2.733	0.006	Supported	0.112 (Medium)
H4: PEU $\rightarrow$ BIU	0.258	2.789	0.005	Supported	0.108 (Medium)
H5: PT $\rightarrow$ BIU	0.299	2.657	0.008	Supported	0.125 (Medium)

The results directly answer RQ1, confirming that both PU and PEU are significant positive drivers of BIU. Specifically, PU ($\beta = 0.281$, $p < 0.01$) and PEU ($\beta = 0.258$, $p < 0.01$) directly influence the intention to adopt blockchain. These findings align with the core tenets of TAM, suggesting that even in bureaucratic settings, perceived utility and ease of use remain fundamental catalysts for adoption intention.

Furthermore, H1 and H2 reveal that functional perceptions significantly build trust. PU exerts the strongest effect on PT ($\beta = 0.548$, $f^2 = 0.365$), followed by PEU ($\beta = 0.242$). This empirically supports the argument based on Attribution Theory: in the high-risk context of fine sanctions, civil servants attribute credibility and trust to the blockchain system primarily when they perceive it as

functionally beneficial (transparent and traceable) (Belanche et al., 2012). H5 is also supported, validating that PT ($\beta = 0.299$) is a crucial predictor of BIU, consistent with the uncertainty reduction logic that trust mitigates perceived risks in digital governance (Dwi & Aljoza, 2015; Rahardja et al., 2023).

Mediation Effects (Addressing RQ2 / RO2)

RQ2 investigates the mediating role of PT in the relationship between TAM constructs (PU, PEU) and BIU. To answer this, specific indirect effects were calculated using bootstrapping.

As detailed in Table 7, the indirect effects of $PU \rightarrow PT \rightarrow BIU$ and $PEU \rightarrow PT \rightarrow BIU$ are both statistically significant. Because the direct paths (H3 and H4) remain significant while the indirect paths are also significant, partial mediation is established.

Table 7. Mediation analysis (specific indirect effects)

Hypothesis	Indirect Path	Coefficient (β)	T-Statistic	P-Value	Mediation Type
H6	$PU \rightarrow PT \rightarrow BIU$	0.164	2.413	0.016	Partial mediation
H7	$PEU \rightarrow PT \rightarrow BIU$	0.072	1.986	0.047	Partial mediation

These results conclusively answer RQ2. PT operates as a vital cognitive filter. The utility (PU) and convenience (PEU) of the blockchain system do not solely drive adoption intention directly; they must also be internalized as system trust (PT) to effect behavioral change. This finding confirms the TGTA mechanism proposed in the literature review, underscoring that in institutional complexity, trust acts as a psychological bridge converting utilitarian evaluation into adoption intention.

Explanatory Power and Predictive Relevance

The model exhibits robust explanatory power. The R^2 value for PT is 0.533, indicating that PU and PEU explain 53.3% of the variance in trust formation. The R^2 value for BIU is 0.542, meaning the model accounts for 54.2% of the variance in adoption intention (Table 8). According to PLS-SEM criteria (Ramalho et al., 2019), both values are classified as substantial. Additionally, the Q^2 values (calculated via blindfolding) for PT and BIU were greater than zero, confirming the model's out-of-sample predictive relevance for policy formulation. **Why Q^2 values are not shown in the table?**

Table 8. R-square Assessment

	R Square	R Adjusted Square
BIU	0.542	0.533
PT	0.533	0.527

DISCUSSION

This study sets out to unravel the behavioral intention to adopt blockchain technology in the governance of telecommunications administrative fines, specifically addressing the gaps of trust decontextualization and empirical deficiency in developing economies. The empirical findings,

derived from the hybrid TAM-TPB-Trust model, provide nuanced insights that challenge traditional linear adoption paradigms and validate the proposed Trust-Guided Technology Assimilation mechanism.

The Primacy of Utility in Institutional Trust Building

The most salient finding of this study is the dominant effect of PU on PT ($\beta = 0.548$, $f^2 = 0.365$), which stands as the strongest pathway in the structural model. This refutes the conventional application of TAM—which typically positions PU merely as a direct driver of usage—and asserts that in high-risk public sector environments, utility is a prerequisite for trust. This aligns with the logic of Attribution Theory, where civil servants attribute credibility to a system only when its functional benefits (e.g., traceability and fraud mitigation) are visibly demonstrated.

This finding resonates with prior e-government adoption studies suggesting that trust in government technology is largely performance-based (Belanche et al., 2012). However, our study extends this by contextualizing it within DLT. Unlike commercial applications where trust is often built through brand reputation or user reviews, blockchain in sanction enforcement is a mandate. Therefore, trust must be engineered through utility. Our data corroborates this: civil servants will not trust an immutable ledger to manage sensitive PNB (non-tax state revenue) merely because it is technologically secure; they trust it because they perceive it as functionally superior in reducing bureaucratic friction and human error.

The Dual Role of Perceived Ease of Use: Cognitive Load and Reliability

While PU dominates trust formation, PEU also significantly influences both PT ($\beta = 0.242$) and BIU ($\beta = 0.258$). In the context of institutional complexity, PEU operates dually. First, based on Cognitive Load Theory, a user-friendly blockchain interface reduces the psychological intimidation associated with transitioning from manual ledgers to cryptographic systems, thereby directly encouraging adoption intention. Second, PEU enhances trust, albeit less robustly than PU. This suggests that while a seamless user experience signals system reliability and reduces the fear of operational errors, ease of use alone cannot compensate for a lack of tangible functional benefits. This nuanced finding supports [the Unified Theory of Acceptance and Use of Technology \(UTAUT\)](#), where effort expectancy is crucial in the early stages of technology exposure but is eventually overshadowed by performance expectancy as users seek actual value.

Trust as the Cognitive Filter: Validating the TGTA Mechanism

Addressing RQ2, the mediation analysis confirms that PT acts as a critical cognitive filter, partially mediating the relationship between TAM constructs (PU, PEU) and BIU. This finding provides empirical validation for the TGTA mechanism, addressing the theoretical gap of trust decontextualization. Traditional TAM assumes a somewhat direct translation of perceived benefits into behavioral intent. However, in the enforcement of administrative fines—a domain highly susceptible to disputes and audit scrutiny—this translation is not automatic.

The partial mediation indicates that while PU and PEU have direct leverage on adoption intention, their influence is significantly amplified when internalized as trust. This finding bridges individual psychological processing and institutional realities. Drawing upon Luhmann's (1979) theory of trust as a mechanism for reducing social complexity, we argue that civil servants utilize PT to bridge the gap between the known (current manual systems) and the unknown (decentralized blockchain). PU provides the rational justification for adoption, but PT provides the psychological safety to commit to it. This extends recent literature on blockchain adoption in supply chains (Bano & Siddiqui, 2025) by demonstrating that in public governance, trust is not just an antecedent or an outcome, but a pivotal translational mechanism.

Contextualizing within Institutional Complexity

Finally, the demographic reality that 78% of respondents have over a decade of work experience and operate within rigid hierarchies (Heads of Office, Law Officers) contextualizes these findings deeply.

In such settings, institutional inertia is profound. The significant direct effect of PT on BIU ($\beta = 0.299$) underscores that in developing economies like Indonesia, technological solutions must be preceded by institutional trust-building. Blockchain promises "trust-less" systems (trust in code rather than humans), yet paradoxically, the adoption of such systems requires high institutional trust from the users toward the proposing authority. The transition from manual oversight to trust-minimized code represents a paradigm shift that civil servants will only accept if they trust the system's integrity (data immutability) and its alignment with their operational safety.

Therefore, embedding TGTA into policy design is pivotal. As Indonesia accelerates its digital transformation, Komdigi must recognize that deploying blockchain is insufficient; the deployment strategy must explicitly demonstrate how the technology solves specific bureaucratic pain points (enhancing PU) and simplify workflows (enhancing PEU) to organically cultivate the trust necessary for sustainable adoption.

CONCLUSION

This study unravels the intricate behavioral dynamics underlying blockchain technology adoption in the governance of administrative fine sanctions within Indonesia's telecommunications sector. By synthesizing the TAM and TPB through the lens of Perceived Trust (PT), this research moves beyond traditional, linear technology acceptance models to address the critical gaps of trust decontextualization and empirical deficiency in developing digital economies.

Our empirical findings confirm that the transition to blockchain in high-risk public sector environments is not merely a technical upgrade, but a psychological reconstruction of institutional confidence. The central tenet of this study—the Trust-Guided Technology Assimilation (TGTA) mechanism—is empirically validated. We demonstrate that Perceived Trust acts as an indispensable cognitive filter; the functional benefits of blockchain (Perceived Usefulness) and its ergonomic design (Perceived Ease of Use) must be internalized as system trust before they can effectively translate into Behavioral Intention to Use (BIU). Notably, the dominant effect of PU on PT ($\beta = 0.548$) reveals a crucial paradigm: in the enforcement of sensitive state revenues, utility is the primary engine of trust. Civil servants will not trust an immutable ledger simply because it is cryptographically secure; they trust it because they perceive it as functionally superior in mitigating bureaucratic friction and fraud.

Theoretical Implications

This study makes three pivotal contributions to the information systems literature. First, it resolves the decontextualization of trust by positioning PT as a partial mediator rather than a parallel construct, proving that in governance settings, trust bridges the gap between rational evaluation (utility) and behavioral commitment. Second, it addresses institutional complexity by demonstrating that standard TAM assumptions require contextual calibration; in rigid bureaucratic hierarchies, utility holds significantly more weight than ease of use in forging trust. Third, it provides the first empirical benchmark for DLT adoption readiness among public service personnel in the ASEAN region, addressing the glaring empirical deficit in the literature concerning non-financial public sector governance.

Practical Implications

For policymakers and the Ministry of Communication and Digital (Komdigi), the findings prescribe a clear strategic roadmap. The paradox of blockchain—a "trust-less" technological system that paradoxically requires high institutional trust to be adopted—demands a shift in implementation strategy. Simply mandating the technology or providing technical training is insufficient. Komdigi must prioritize utility demonstration: pilot programs must explicitly showcase how the DLT system reduces procedural errors, secures sensitive data, and protects civil servants from audit liabilities. Building PT requires transparent communication about data immutability and system reliability. By

first securing user trust through visible functional benefits, the resistance inherent in institutional complexity can be systematically dismantled, paving the way for sustainable digital transformation.

LIMITATIONS AND FUTURE RESEARCH

First, because the data is cross-sectional, the causal relationships inferred capture only a snapshot in time and may not reflect long-term trust dynamics as users gain actual hands-on experience with the DLT system. While providing robust insights, this study has limitations that offer avenues for future research. As blockchain initiatives in Indonesia mature, longitudinal studies are needed to assess the transition from intention to actual usage. Second, this study focuses on the cognitive-psychological constructs of TAM-TPB-Trust; future research could enrich the model by incorporating organizational variables, such as top management support or organizational culture, which may further explain the variance in adoption within complex bureaucracies. Finally, while the telecommunications sector provides a stringent empirical context, comparative studies across different public service domains (e.g., public health or land registries) would enhance the generalizability of the TGTA mechanism.

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